

VISHNU RAMU

Application Engineer

Chennai, Tamil Nadu

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Professional Summary

Motivated and detail-oriented Machine Vision Application Engineer with hands-on experience in designing, developing, and deploying 2D/3D vision-based systems. Skilled in working with Mech-Mind 3D vision, Hikrobot's Vision Master software, PLCs, and robotics integration. Strong background in industrial automation, algorithm development, and robotics simulations. Passionate about leveraging technology to improve operational efficiency and precision.

Technical Skills

- Machine Vision: Mech-Mind 3D, Vision Master, HIKROBOT
- Automation Tools: PLC (Siemens, Allen-Bradley), Profinet, Modbus, TCP/IP, SLMP.
- Programming: Python, Arduino IDE, Ladder Logic
- Robotics: ABB RobotStudio, Epson RC+, ROS, Mitsubishi, Yaskawa, KUKA, Doosan, UR, Aubo
- Other: Communication Protocols, Troubleshooting, Simulation, Project Commissioning, Quality Assurance

Professional Experience

Application Engineer

Robotronix Automation India Pvt Ltd – Chennai

May 2024 – Present

- Designed and deployed machine vision systems using Mech-Mind 3D cameras and Hikrobot's 2D cameras and Vision Master software for inspection, alignment, and robotic pick-and-place applications.
- Integrated robot arms (Mitsubishi, Yaskawa) and PLCs (Siemens, Allen-Bradley) with vision systems for industrial automation tasks.
- Executed multiple projects including:
 1. **Vision inspection of mosquito repellent bottles** for defects like bottle dents, wick breakage, liquid level, and top wick damage.
 2. **Tread length measurement** of tyre up to 2.5 meters with 1mm accuracy using vision camera.

3. **Microchip pick and place** using Mitsubishi robot with 20 micrometre accuracy.
 4. **Code reading and angle detection on tyre**, integrated with Allen-Bradley PLC.
 5. **Clip and foam presence/absence inspection** in car bonnets using line scan camera (2m components).
 6. **Height inspection of 16 pins in semiconductor industry** using Mech-Mind 3D profiler.
 7. **Bin picking of automobile components with Yaskawa robot** using Mech-Mind 3D vision.
- Performed calibration, system setup, testing, and on-site support for client installations.
 - Optimized object detection algorithms and system performance based on real-world feedback.

Internship Trainee

Axis Global Automation – Chennai
Feb 2022 – Mar 2023

- Assisted in configuring and programming PLC and SCADA systems for industrial automation projects.
- Supported the setup, testing, and debugging of automation sequences.
- Gained hands-on experience with industry-standard tools and practices.

Student Intern

SmartKnower – Remote
Sep 2022 – Oct 2022

- Simulated a pick and place robotic arm using ROS and Arduino.
- Learned basics of robotics simulation and control system logic.

Education

Master of Technology – Artificial Intelligence
SRM University - Kattankulathur
March 2025 – May 2027

Bachelor of Engineering – Robotics and Automation Engineering
Sengundhar College of Engineering, Namakkal
June 2019 – June 2023

Certifications

- Mech-Mind 3D Vision-Guided Picking - Beginner
- Six Sigma White Belt
- Voice App Training in Bolt IoT
- Robotics Process Automation with UiPath
- Best Starter Course: Robotics & ROS
- Electric Vehicles in E-mobility

Academic Projects

Brain-sense Based Wheelchair for Differently Abled

Developed a wheelchair controlled by EEG signals via Arduino to help differently abled individuals navigate using thoughts.

Stair Climbing Robot using Rocker-Bogie Mechanism

Designed and fabricated a robot that climbs stairs with stability and control.

Smart Security System with Infrared Sensor

Built and deployed a real-time IR and GSM-based security system installed in a supermarket.

Fastest Line Follower Robot with IR Sensor

Won first prize among 20 entries for developing a high-speed line follower robot.

Pick and Place Robot Cell Simulation (ABB RobotStudio)

Simulated a conveyor-integrated robotic pick and place system using ABB RobotStudio.

Languages

- English (Fluent)
- Tamil (Native)